

Session 2 · Deck 2.A1

Bewley & Huggett: Why Can't Everyone Insure?

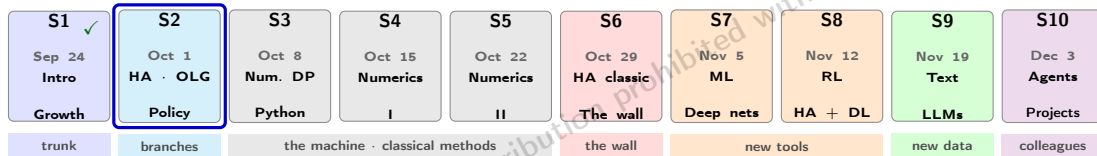
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Thursday, October 1, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

▼ you are here: 2.A1

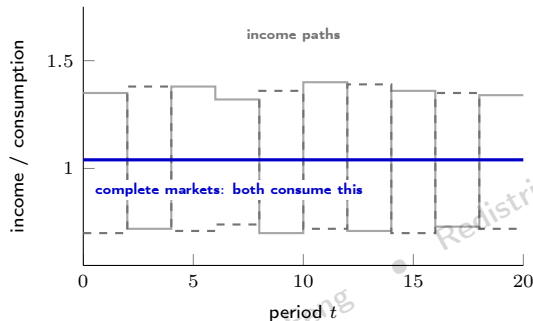


Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 2.A1 why insurance fails and what replaces it · **2.A2** Aiyagari: production and the capital market · **2.A3** the stationary distribution, and where the model is taken next.

A Prediction You Can Test

Two households, same lifetime income, different luck



Thin lines: two realizations of the same i.i.d. endowment process. Thick line: the full-insurance allocation.

The complete-markets prediction

With a full set of state-contingent claims, marginal utilities move *together*:

$$\frac{u'(c_{t+1}^i)}{u'(c_t^i)} = \frac{u'(c_{t+1}^j)}{u'(c_t^j)} \quad \text{for all } i, j.$$

Individual income is **irrelevant** for consumption growth; only aggregate risk is priced.

It fails in the data. Consumption responds to idiosyncratic income — more so for the young and the poor.

Cochrane (1991); Attanasio & Davis (1996); Blundell, Pistaferri & Preston (2008).

Where the Insurance Goes Missing

The Arrow–Debreu benchmark. Trade at date 0 in claims to every history s^t . With planner weights $\{\theta^i\}$,

$$u'(c^i(s^t)) = \frac{1}{\theta^i} \mu(s^t),$$

one multiplier for everyone: individual consumption is a fixed function of *aggregate* resources.

Take that away. Three frictions, any one of which suffices:

- **Missing markets** — only a risk-free claim trades.
- **Limited commitment** — contracts cannot be enforced.
- **Private information** — the shock is unobserved.

This course takes the first road

We *assume* the asset menu: one risk-free bond, plus a borrowing limit. This is the **Bewley–Huggett–Aiyagari** tradition.

It is assumed, not derived, because it is tractable, disciplined by data, and computable.

Research in Practice

The friction is a *modelling choice*, not a theorem. A welfare number from such a model inherits that choice — state it when you report one.

Huggett (1993): The Simplest Economy That Works

- A continuum of households, unit mass. **No production:** each receives an endowment $e_t \in \mathcal{E}$, a finite Markov chain with transition $\pi(e, e')$.
- One asset: a one-period discount bond, price q , paying one unit next period. Holdings a , with $a' \geq -\underline{a}$.
- The household solves

$$v(a, e) = \max_{c, a'} \left\{ u(c) + \beta \sum_{e'} v(a', e') \pi(e, e') \right\}$$

subject to $c + q a' = a + e$, $a' \geq -\underline{a}$, $c \geq 0$.

- **Zero net supply:** the bond is an IOU between households. In equilibrium

$$\int a'(a, e) d\lambda(a, e) = 0.$$

Why start here

No capital, no firms, no wage. The **only** price is q , and the only equilibrium condition that bites is that borrowing equals lending.

Everything hard about heterogeneous agents is already present: a cross-sectional distribution λ , a price that depends on it, and no planner to appeal to.

$\lambda(a, e)$ is a **probability measure** over (a, e) , not a list of names. Households are anonymous; only the distribution matters for prices.

The Borrowing Limit Is Not a Free Parameter

Natural limit. The largest debt a household can *always* repay, even drawing the worst endowment forever:

$$\underline{a}^{\text{nat}} = \frac{y_{\min}}{r},$$

which in the Aiyagari economy of 2.A2 is $w\epsilon_{\min}/r$.

Ad hoc limit. Any $\underline{a} \leq \underline{a}^{\text{nat}}$, imposed to match observed debt or to stand in for unmodelled default risk.

- At the natural limit it never binds — $c \rightarrow 0$ is avoided only because $u'(0) = \infty$.
- Tighter limits make it bind for a positive mass.
That mass is what policy questions care about.

Always $a' \geq -\underline{a}$ with $\underline{a} \geq 0$ here; the sister-course slides flip this sign.

A modelling decision with teeth

The constrained fraction drives the consumption response to a transfer and the optimal level of social insurance. In most papers it is chosen, not estimated. Say which you did.

Or derived, not assumed

Take the other two roads and the limit becomes an **equilibrium object**. **The cost is computational:** no planner's problem, and a recursive equilibrium on (a, e) may not exist — you need an **expanded state space** whose fixed point is the whole set of equilibria. Kehoe–Levine (1993); Alvarez–Jermann (2000); Feng, Miao, Peralta-Alva & Santos (IER 2014) — 1.A2's warning made concrete; the device returns in 2.B3.

The Household's Optimality Conditions

Attach a multiplier η to the borrowing constraint. The **Kuhn–Tucker** conditions of the household's problem are the Euler equation

$$q u'(c) - \beta \mathbb{E}[u'(c') | e] = \eta$$

together with **complementary slackness**,

$$\eta \geq 0, \quad a' + \underline{a} \geq 0, \quad \eta (a' + \underline{a}) = 0.$$

Those three conditions split the problem in two:

- $a' > -\underline{a} \Rightarrow \eta = 0$: the Euler equation holds **with equality**.
- $a' = -\underline{a} \Rightarrow \eta > 0$: $q u'(c) > \beta \mathbb{E}[u'(c') | e]$ — the household *would* borrow more, and cannot.

u is concave on a convex budget set, so these are necessary **and** sufficient.

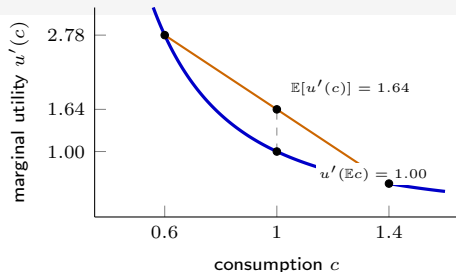
η is what the solver chases

The multiplier is not a bookkeeping device. It is the **shadow price of the borrowing limit** — how much a household would pay to be allowed one more unit of debt.

It is zero for most households and strictly positive for the ones every policy question is about.

And it is why the problem is awkward to compute: the system is a set of *inequalities*, and which branch applies is itself unknown.

Precautionary Saving: Where the Extra Assets Come From



$u'(c) = c^{-2}$; $c \in \{0.6, 1.4\}$ equally likely, so $E[c] = 1$. The chord lies

above the curve: Jensen.

Why risk raises saving. Marginal utility is **convex**, so a mean-preserving spread raises its expectation:

$$E[u'(c')] = 1.64 > 1.00 = u'(E[c]).$$

The right-hand side of the Euler equation is bigger than under certainty, so the household saves more at every r .

Invert it: the household behaves as though consumption were $1.64^{-1/2} = 0.78$ rather than 1.00. That gap of 0.22 is the **precautionary premium**, and the extra assets are what buys it.

Prudence, not risk aversion, and two motives, not one. Convexity of u' is $u''' > 0$: quadratic utility is risk averse, has $u''' = 0$, and gets *no* precautionary saving at all (CRRA has relative prudence $\gamma + 1$ against risk aversion γ). Prudence operates even where the limit is never reached; a limit that may bind *later* adds a **second** motive, which bites even where $\eta = 0$ today. Leland (1968); Kimball (1990); Carroll & Kimball (1996). What the buffer is for is three slides on.

One Equation Instead of Three Conditions

Those three conditions are a **complementarity problem**:

$$0 \leq \eta \perp (a' + \underline{a}) \geq 0.$$

A root-finder wants *equations* and a gradient wants *smoothness*; inequalities give neither. The **Fischer–Burmeister** function turns all three into one:

$$\phi(x, y) = x + y - \sqrt{x^2 + y^2},$$

which vanishes *exactly* when $x \geq 0$, $y \geq 0$ and $xy = 0$. So $\phi(\eta, a' + \underline{a}) = 0$ makes the household problem a square system for Newton's method.

Three ways this is handled

1. Solve the unconstrained Euler equation, then set $a' = \max\{-\underline{a}, a'_{\text{unc}}\}$ (S6).
2. FB as an equation inside a Newton solve (S4).
3. FB as a *penalty* $\lambda \sum \phi_\varepsilon(\cdot)^2$ — how a neural solver enforces what it cannot impose (S8).

$\phi_\varepsilon = x + y - \sqrt{x^2 + y^2 + \varepsilon^2}$ rounds the kink at the origin. ε too large biases the answer off the constraint; too small kills the Newton step. It is a choice you report.

The Income Process Is the Model's Engine

Everything the model says about wealth runs through $\pi(e, e')$. The standard specification is a discretized AR(1) in logs,

$$\log e_{t+1} = \rho \log e_t + \sigma \varepsilon_{t+1}, \quad \varepsilon \sim N(0, 1),$$

turned into a finite Markov chain (Tauchen 1986; Rouwenhorst 1995 — we build these in S4).

- ρ **governs how long a bad draw lasts**. U.S. earnings estimates put ρ near 0.95 annually.
- σ **governs how bad it gets**.
- Together they set how much wealth a household *needs* to self-insure — and therefore the whole shape of λ .

Persistence is not a detail

Under **i.i.d.** shocks a bad draw lasts one period, so a modest buffer suffices and almost nobody is ever driven to the constraint.

Under **persistent** shocks a bad spell can last years, the required buffer is large, and a real mass of households ends up constrained.

Same equilibrium concept, same code, radically different economics. Persistence is a parameter *you* choose, and it decides whether the model has any constrained households to study. **We put a number on it in S6.**

How Good Is Self-Insurance?

Assets are a **buffer stock**, not a contract: they insure by *substituting across time*, and unevenly.

- **Rich households** are nearly fully insured — a bad draw is absorbed by running down assets.
- **Households near the constraint** are barely insured: consumption tracks income one-for-one.
- So the **MPC** out of a transfer is near 0 at the top and near 1 at the bottom — heterogeneity a representative agent cannot have, and the reason these models are used for fiscal policy.

Two planners, two failures

An **unconstrained** planner allocates directly, so it *would not choose* this market structure — it is an assumption of the model, not a constraint on the planner.

A **constrained** planner, held to the same asset, can usually *still* improve: households ignore how their saving moves r . Dávila, Hong,

Krusell & Ríos-Rull (2012).

Research in Practice

So we solve the **equilibrium** directly — and in 2.A3, assets are what let a household absorb a medical bill or fund a business.

Stationary Equilibrium in the Huggett Economy

Definition

A **stationary recursive competitive equilibrium** is a bond price q^* , a value function v , policies (c, a') , and a measure λ^* on $\mathcal{A} \times \mathcal{E}$ such that

1. given q^* , (v, c, a') solve the household problem;
2. the bond market clears: $\int a'(a, e) d\lambda^*(a, e) = 0$;
3. λ^* is invariant under the transition induced by $a'(\cdot)$ and π :

$$\lambda^*(\mathcal{B}, e') = \sum_e \pi(e, e') \int_{\{a : a'(a, e) \in \mathcal{B}\}} d\lambda^*(a, e).$$

Why *this* equilibrium?

Stationary — λ constant, prices are *numbers* (2.A, S6).

Recursive / global — Γ moves, prices are *functions* of Γ (2.B1, S6–S8).

Transition path — a sequence $\{\lambda_t\}$ (S6).

Only in the first does the distribution **drop out of the household's state**. Huggett's own reason for stopping there: in 1993 “general methods for characterizing equilibria of the more general kind do not currently exist.”

Why the Interest Rate Must Be Below ρ

The argument, in three steps.

1. Suppose $q = \beta$, i.e. $r = \rho$. Then the Euler equation reads $u'(c_t) = \mathbb{E}_t[u'(c_{t+1})] + \eta_t$, so $\{u'(c_t)\}$ is a **non-negative supermartingale**.
2. A non-negative supermartingale converges almost surely. But with a persistent, non-degenerate endowment and a finite debt limit, $u'(c_t)$ cannot settle down unless assets grow without bound.
3. Every household therefore wants to accumulate forever: aggregate asset demand is $+\infty$. That cannot clear a market in **zero net supply**.

Hence $q^* > \beta$, i.e. $r^* < \rho$.

Read it as a price

A low r is the price that **chokes off** precautionary demand until borrowing equals lending. The more risk, or the tighter the limit, the lower r^* must go.

Research in Practice

The incomplete-markets answer to the risk-free rate puzzle — and why these models are used for public debt: the government borrows cheaply precisely because households cannot insure.

Bewley (1986); Huggett (1993); Aiyagari (1994); Chamberlain & Wilson (2000).

Does a Stationary Equilibrium Exist?

The half that is a theorem

Fix q . If

1. $e_{\min} - (1 - q)\underline{a} > 0$ — at the debt limit, drawing the worst endowment for ever, consumption is still *positive*;
2. $\beta < q$, that is $r < \rho$ — the previous slide;
3. $\pi(e_h | e_h) \geq \pi(e_h | e_l)$ — the shock chain is *monotone*;

then there is a **unique** stationary measure $\lambda(q)$, and iterating any initial distribution forward converges weakly to it — from **any** starting point.

Huggett (1993), Thm. 2, via Hopenhayn & Prescott (1987), Thm. 2 — the *monotone mixing condition*. Those three hypotheses are the three properties of 2.A3, bundled.

The half that is not

None of that delivers the *price*. Huggett is explicit:

"It is not known whether stationary equilibria are unique. If the excess demand for credit balances is continuous and strictly decreasing in the price ... the intermediate value theorem would give the existence of a unique equilibria. These properties have been difficult to prove."

Excess demand was strictly decreasing in every case he computed. The gap is still open — and reappears for Aiyagari's r^* in 2.A3.

What the Model Delivers, and What It Does Not

Delivers

- A non-degenerate **wealth distribution** from identical households and luck alone.
- **Social mobility**, at a rate set by π .
- A risk-free rate **below** the rate of time preference.
- A welfare cost of idiosyncratic risk.
- A stationary distribution that **exists, is unique, and is reached from any start** (Huggett) — and in 2.A2 you watch it happen, in Feng, Han, Sargent & Zhu (2026).

Does not deliver

- The **top tail**. Realistic earnings risk produces nothing like the top 1% wealth share — numbers in 2.A3.
- A theory of *why* markets are incomplete.
- Any role for aggregate risk — prices are constant by construction (2.B1).
- Finite lives, bequests, retirement (2.B2).

- Full insurance is a **testable** prediction about consumption growth. It is rejected.
- Assume one asset and a limit and you get precautionary saving, a wealth distribution and $r^* < \rho$ from identical households — with the equilibrium object a **measure** that must reproduce itself.

Bridge: Put a Firm in the Room

Huggett's bond is in **zero net supply**: one household's asset is another's debt, and the interest rate clears that internal market.

Now let the asset be a claim on **physical capital**.

Then:

- aggregate saving is no longer zero — it is the capital stock K ;
- K has a **price determined by technology**, through the firm's first-order condition;
- the wage appears, so the endowment becomes labour income $w\epsilon$.

Next (2.A2): Aiyagari

The same household problem, closed with a neoclassical firm. Equilibrium becomes the crossing of a **capital demand** curve from the firm and a **capital supply** curve from the distribution of households.

And the first quantitative disappointment of the course.